Nonstationarity-Aware Mamba for Time Series Forecasting

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Abstract

Nonstationary time series forecasting is difficult because the data-generating process can change across time. Changes in level, volatility, trend, and temporal regime can make a model trained on past windows unreliable for future horizons. This manuscript studies nonstationarity-aware forecasting with Mamba-based sequence models, using daily GLD close-price prediction as the initial testbed. The proposed model family, NS-Mamba, extends an S_Mamba backbone with adaptive normalization, series decomposition, and distribution-shift-aware fusion. Before evaluating the proposed architecture, we define a controlled baseline protocol across 12 horizons from 1 to 360 days and lock a shared configuration through S_Mamba calibration. The completed GLD baseline results show that S_Mamba is the strongest model from 1 to 90 days on RMSE, while iTransformer is strongest from 120 to 360 days. The first NS-Mamba diagnostic ablation shows that the decomposition-only variant improves long-horizon RMSE over both S_Mamba and iTransformer, but the full 12-horizon NS-Mamba evaluation remains pending. The completed source-optimized imported-model sweep further shows that PatchTST is the strongest imported baseline overall when its own hyperparameters are tuned rather than forced into the shared setting.

1 Introduction

Time series forecasting models are often evaluated under the assumption that past and future windows are drawn from similar distributions. This assumption is weak for nonstationary real-world data. Gold-related exchange-traded products such as GLD can experience abrupt level shifts, volatility changes, macroeconomic regime changes, and long periods of trend persistence or reversal. These effects are not just noise; they change the forecasting problem itself.

Recent deep forecasting models use attention, decomposition, and state-space mechanisms to model long temporal context. Mamba is especially attractive because it provides linear-time sequence modeling through selective state spaces [1]. In a forecasting setting, however, using Mamba alone does not guarantee robustness to nonstationarity. The model may still entangle level, scale, trend, and residual dynamics in one hidden representation. This motivates an architecture that keeps the efficient Mamba backbone while explicitly exposing nonstationary structure to the network.

This work begins with a strict experimental principle. Before introducing NS-Mamba, all suitable baselines are evaluated under a shared configuration selected by S_Mamba calibration. This prevents model-specific tuning from becoming a hidden source of performance differences. The same horizon suite and output scale are then used for clean baselines, Mamba variants, and the proposed NS-Mamba models.

The intended contributions are:

- a controlled GLD forecasting protocol covering horizons from 1 to 360 days;

- a clean comparison between Transformer-style, decomposition-based, inverted, and Mamba-based forecasting models;
- a nonstationarity-aware Mamba architecture family combining adaptive normalization, decomposition, and shift-aware fusion;
- a diagnostic ablation showing that decomposition is the strongest first modification on GLD long-horizon forecasting.

2 Related Work

2.1 Long-Horizon Time Series Forecasting

Long-horizon time series forecasting aims to predict many future steps from a historical lookback window. It is important in domains where decisions must be made before the target is observed, including finance, energy, traffic, weather, and operations planning. Compared with short-term forecasting, long-horizon forecasting is more sensitive to error accumulation, distribution shift, trend changes, and the difficulty of preserving useful long-range temporal information. In financial forecasting, these challenges are amplified because the target can move under changing macroeconomic conditions, market expectations, and volatility regimes. The GLD task studied in this manuscript follows this setting by evaluating horizons from 1 day to 360 days.

2.2 Transformer-Based Time Series Forecasting

Transformer-based forecasting models introduced attention mechanisms for modeling long temporal context. Informer improves efficiency by reducing the cost of attention for long sequences [10]. Reformer reduces attention complexity through locality-sensitive hashing and reversible layers [12]. Autoformer introduces decomposition into the forecasting architecture, separating trend and seasonal components to improve long-term prediction [11]. These models provide important baselines because they represent different design choices for efficient long-context modeling, decomposition, and temporal dependency extraction.

2.3 Inverted Forecasting Architectures

iTransformer argues that treating variables as tokens and encoding their temporal histories can be effective for multivariate forecasting [9]. The S_Mamba baseline in this repository follows a related inverted-token view by embedding each variable's lookback history before applying Mamba encoder layers. For the initial GLD setting, the task is single-target forecasting, but the same design remains relevant because it controls how temporal history is mapped into hidden features.

2.4 State Space Models and Mamba

State space models provide an alternative to attention for sequence modeling. The continuous-time form of a linear state space model is:

$$\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \qquad \mathbf{y}(t) = \mathbf{C}\mathbf{h}(t), \tag{1}$$

where $\mathbf{h}(t)$ is the latent state, $\mathbf{x}(t)$ is the input, and $\mathbf{y}(t)$ is the output. After discretization with step size Δ, the recurrence can be written as:

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{x}_t, \qquad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t, \tag{2}$$

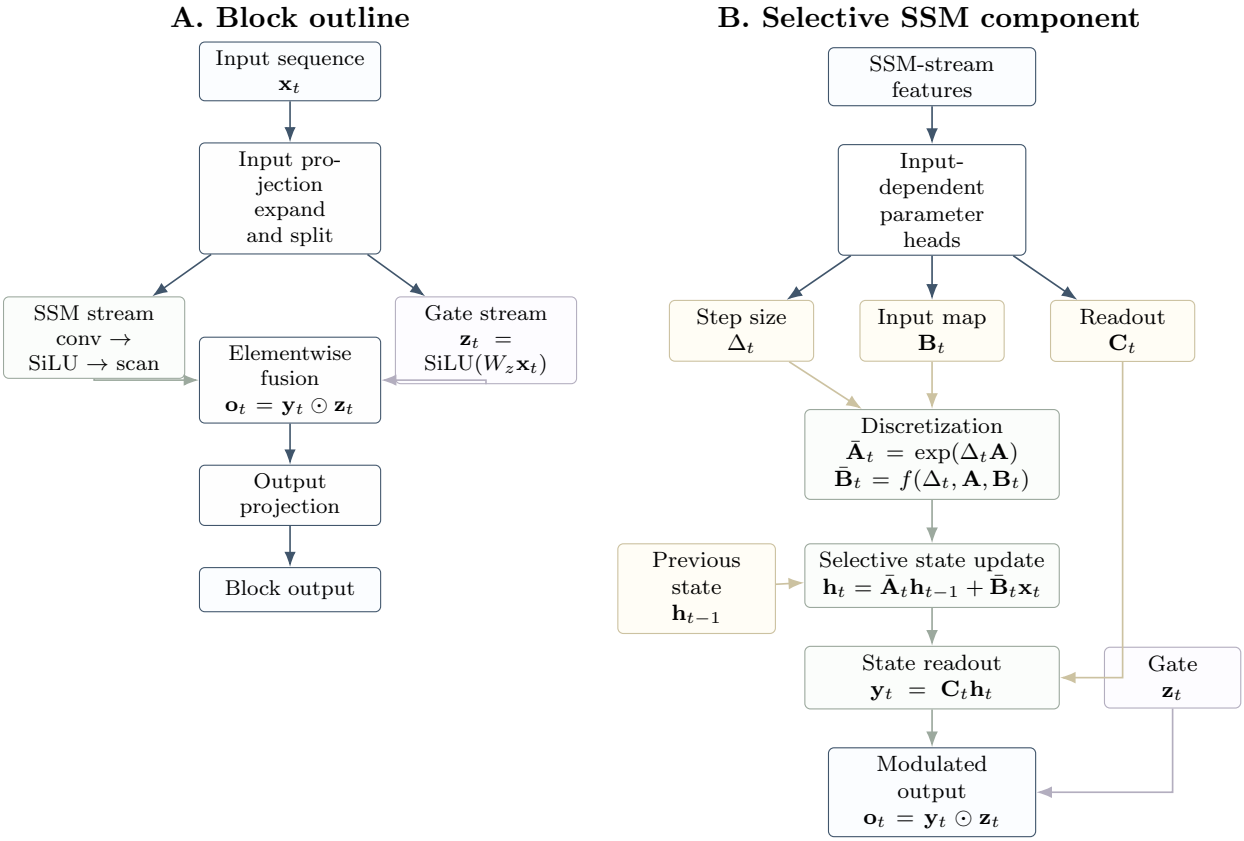


Figure 1: Mamba block architecture drawn with native TikZ so mathematical notation is rendered by LaTeX. Panel A shows the high-level block outline: input projection, SSM stream, gating stream, fusion, and output projection. Panel B expands the key selective state-space component, where input-dependent parameters generate the discretized update, readout, and gated output.

with

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \qquad \bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right)\Delta\mathbf{B}. \tag{3}$$

Classical structured state space models use fixed transition parameters after training. Mamba introduces selective state spaces, where Δ_t, $\mathbf{B}_t$, and $\mathbf{C}_t$ depend on the current input token [1]. The input-dependent recurrence is:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t\mathbf{h}_{t-1} + \bar{\mathbf{B}}_t\mathbf{x}_t, \qquad \mathbf{y}_t = \mathbf{C}_t\mathbf{h}_t, \tag{4}$$

where $\bar{\mathbf{A}}_t$ and $\bar{\mathbf{B}}_t$ are computed from the input-dependent step size and input projection. This selectivity is useful for forecasting because the model can adapt what information is retained across time. Figure 1 summarizes the Mamba block used as the conceptual backbone in this manuscript. The central question is whether explicit nonstationarity modules can make this selective mechanism more effective for data whose distribution changes across windows.

2.5 Mamba-Based Time Series Forecasting

Several recent studies have adapted Mamba to time series forecasting. S_Mamba studies whether Mamba is effective for time series forecasting and uses a bidirectional Mamba layer to model inter-variate correlations with a feed-forward layer for temporal dependencies [2]. Bi-Mamba+ extends

this direction with forward and backward Mamba processing, a forget gate, and a relation-aware choice between channel-independent and channel-mixing tokenization [3]. MambaTS analyzes limitations of directly applying Mamba to long-term forecasting and proposes temporal Mamba blocks, variable scanning strategies, and variable-aware scan ordering [4]. TimeMachine uses multiple Mamba pathways to handle channel-mixing and channel-independence settings while preserving linear scalability for long-term forecasting [5]. TSMamba moves from task-specific forecasting toward a Mamba-based foundation model, using forward and backward Mamba encoders and transfer learning for time series prediction [6]. Together, these examples show that Mamba has become an active backbone for forecasting, but they do not directly resolve the specific problem studied here: explicit nonstationarity handling through adaptive normalization, decomposition, and shift-aware fusion.

2.6 Nonstationarity and Normalization

Distribution shift is a central issue in time series forecasting. RevIN uses reversible instance normalization to reduce distribution shift between training and testing windows while restoring the original scale for prediction [7]. Non-stationary Transformers introduce mechanisms to handle stationarity and nonstationarity within Transformer forecasting [8]. NS-Mamba follows the same broad motivation but studies it in a Mamba-based architecture with adaptive normalization, decomposition, and shift-aware fusion.

3 Problem Formulation and Nonstationarity

Let

$$\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L\times C} \tag{5}$$

denote an input window of length L with C variables. The goal is to predict

$$\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H\times C_o}, \tag{6}$$

where H is the forecast horizon and C_o is the output dimension. For the GLD study, $C = C_o = 1$ and the target is the daily `close` column.

The evaluated horizon set is:

$$H \in \{1, 3, 5, 7, 14, 30, 60, 90, 120, 180, 270, 360\}. \tag{7}$$

These horizons cover short-term, middle-term, and long-term forecasting behavior.

Nonstationarity means that the joint distribution of input and output windows changes with time. Let $\mathcal{D}_t$ denote the local joint distribution around time t. The nonstationary setting can be written as:

$$\mathcal{D}_t(\mathbf{X}_{t-L+1:t}, \mathbf{Y}_{t+1:t+H}) \neq \mathcal{D}_{t+\tau}(\mathbf{X}_{t+\tau-L+1:t+\tau}, \mathbf{Y}_{t+\tau+1:t+\tau+H}). \tag{8}$$

This mismatch means that minimizing empirical training risk does not necessarily minimize future test risk. In GLD forecasting, common sources of mismatch include changing price level, volatility expansion or contraction, trend reversal, and regime-specific temporal dependence.

The proposed approach is based on three assumptions. First, local level and scale variation should be normalized before sequence modeling. Second, trend-like and residual-like movements should not always be forced into the same representation. Third, the model should receive explicit shift statistics so that it can change its fusion behavior under different regimes.

4 Method: NS-Mamba

NS-Mamba is designed as a modular extension of S_Mamba. The base model provides the sequence backbone, while the new modules expose nonstationary structure. Given an input window $\mathbf{X}$, the model computes:

$$\hat{\mathbf{Y}} = \mathrm{DeNorm}\left(F_\theta\left(\mathrm{Decomp}\left(\mathrm{AdaptiveNorm}(\mathbf{X})\right), \phi(\mathbf{X})\right)\right), \tag{9}$$

where F_θ is the Mamba forecasting network and $\phi(\mathbf{X})$ is a vector of distribution-shift statistics.

4.1 Adaptive Normalization

The first module normalizes the input window and learns adaptive affine parameters from window statistics. The normalized input is:

$$\tilde{\mathbf{X}} = \gamma(\phi(\mathbf{X})) \odot \frac{\mathbf{X} - \mu(\mathbf{X})}{\sigma(\mathbf{X}) + \epsilon} + \beta(\phi(\mathbf{X})). \tag{10}$$

Here $\mu(\mathbf{X})$ and $\sigma(\mathbf{X})$ are local window statistics, while $\gamma(\cdot)$ and $\beta(\cdot)$ are learned functions. This module generalizes fixed reversible normalization by allowing the normalized representation to depend on distribution-shift features.

4.2 Series Decomposition

The second module decomposes the normalized window into trend, seasonal, and residual components:

$$\tilde{\mathbf{X}} = \tilde{\mathbf{X}}^{\mathrm{trend}} + \tilde{\mathbf{X}}^{\mathrm{seasonal}} + \tilde{\mathbf{X}}^{\mathrm{residual}}. \tag{11}$$

The initial implementation uses moving-average decomposition. The purpose is not to assert a perfect additive data-generating process, but to provide an architectural bias that separates slow and fast temporal variation.

4.3 Shift-Aware Fusion

The third module uses the shift statistics $\phi(\mathbf{X})$ to control the fusion of hidden representations. For branch outputs $\mathbf{z}_k$, the fusion weights are:

$$\boldsymbol{\alpha} = \mathrm{softmax}\left(\mathbf{W}_\alpha \phi(\mathbf{X}) + \mathbf{b}_\alpha\right), \tag{12}$$

and the fused representation is:

$$\mathbf{z} = \sum_k \alpha_k \mathbf{z}_k. \tag{13}$$

This design allows the model to emphasize trend, seasonal, or residual branches differently when the input window exhibits different shift patterns.

5 Neural Network Architecture

5.1 Input and Tensor Shapes

The implementation uses tensors of shape $\mathbb{R}^{B \times L \times C}$, where B is batch size. For GLD, $C = 1$. After adaptive normalization and optional decomposition, each component is passed to an inverted embedding layer. The embedding maps temporal history into hidden dimension d_{model}. In the locked GLD setting, $d_{\mathrm{model}} = 64$ and the feed-forward dimension is 128.

5.2 S_Mamba Encoder

The S_Mamba backbone applies bidirectional Mamba encoder layers to the embedded representation. A simplified state-space update is:

$$\mathbf{h}_t = \mathbf{A}(\mathbf{x}_t)\mathbf{h}_{t-1} + \mathbf{B}(\mathbf{x}_t)\mathbf{x}_t, \qquad \mathbf{o}_t = \mathbf{C}(\mathbf{x}_t)\mathbf{h}_t. \tag{14}$$

The input-dependent matrices give Mamba a selective memory mechanism. NS-Mamba preserves this mechanism and adds explicit nonstationary conditioning before and around the encoder.

5.3 Branch Design

NS-Mamba-A contains the adaptive normalization block and one S_Mamba forecasting pathway. NS-Mamba-D uses trend-seasonal-residual decomposition with standard S_Mamba normalization. NS-Mamba-AD combines adaptive normalization and decomposition. NS-Mamba-ADG adds shift-aware branch fusion on top of the adaptive decomposition pathway. The current implementation uses component-specific branch encoders for the decomposition variants.

5.4 Forecast Head and Loss

The forecast head projects the final hidden representation to H future values. The training objective is mean squared error over the forecast horizon:

$$\mathcal{L} = \frac{1}{BHC_o} \sum_{b=1}^{B} \sum_{h=1}^{H} \sum_{c=1}^{C_o} \left(\hat{Y}_{b,h,c} - Y_{b,h,c} \right)^2. \tag{15}$$

In the current codebase, inverse transformation is enabled by default, so saved predictions, targets, and metrics are reported on the original GLD price scale unless a run uses `--no-inverse`.

6 Experimental Setup

6.1 Dataset

The initial dataset is GLD daily data stored at `data/GLD/GLD.csv`. The target column is `close`. The first experimental phase uses single-target forecasting. Future extensions may add more financial or macroeconomic variables, but those are outside the current baseline protocol.

6.2 Data Splits

The repository uses the standard custom-data split from the forecasting codebase. The validation and test windows use historical lookback context from the previous split to construct the input sequence. This is a common forecasting implementation detail, but it should be reported clearly because the input context for validation and test includes observations before the split boundary.

6.3 Baseline Models

The core baseline set is:

- S_Mamba;
- Transformer;

- Informer;
- Reformer;
- Autoformer;
- iTransformer.

Optional Mamba-augmented variants will be reported separately if they are included. The clean baselines should not be tuned independently after the shared setting is locked.

6.4 Shared Configuration

S_Mamba calibration selected Setting A as the locked baseline configuration:

Parameter	Value
Lookback length	360
Label length	180
Prediction length	horizon-specific
Model dimension	64
Feed-forward dimension	128
Encoder layers	2
Batch size	16
Target	`close`
Output scale	original price scale

All suitable models will use this configuration. If a model cannot run under this setting, the failure will be recorded rather than changing only that model's hyperparameters.

6.5 Metrics and Logging

The primary manuscript metrics are RMSE and MAE. MSE, MAPE, and MSPE are also saved when available. Training, validation, and test metrics are logged to TensorBoard. Final arrays are saved as `pred.npy`, `true.npy`, and `metrics.npy`, with run metadata stored in `meta.json`. The completed GLD baseline sweep contains 72 finished runs, corresponding to 6 models across 12 horizons.

7 Results

The GLD baseline evaluation is complete under the locked shared setting. All 72 runs finished successfully, corresponding to 6 core models evaluated across 12 horizons. The machine-readable result table is stored in `results/gld_baselines.csv`.

The main pattern is a two-model regime. S_Mamba is the strongest RMSE baseline for 8 horizons, covering every horizon from 1 to 90 days, while iTransformer is strongest for the remaining 4 horizons from 120 to 360 days. The handover is narrow rather than abrupt: at 120 days the RMSE gap is about 2.05% in favor of iTransformer, while at 360 days the gap is only about 0.06%. This means the long-horizon comparison between the two models is highly competitive rather than dominated by one side.

Autoformer is a stable third-place baseline across all 12 horizons. However, its overall average RMSE of 38.71 is far above the two leading models, whose overall average RMSE values are 16.04

Table 1: Best and second-best GLD baseline models by horizon, measured with RMSE.

Horizon	Best model	RMSE	Runner-up	RMSE
1	S_Mamba	4.45	iTransformer	4.59
3	S_Mamba	5.30	iTransformer	5.45
5	S_Mamba	5.76	iTransformer	6.12
7	S_Mamba	6.47	iTransformer	6.85
14	S_Mamba	8.17	iTransformer	8.55
30	S_Mamba	11.12	iTransformer	11.45
60	S_Mamba	14.80	iTransformer	14.89
90	S_Mamba	16.68	iTransformer	16.89
120	iTransformer	19.89	S_Mamba	20.31
180	iTransformer	26.71	S_Mamba	26.91
270	iTransformer	32.95	S_Mamba	33.00
360	iTransformer	39.45	S_Mamba	39.48

Table 2: Average GLD baseline RMSE grouped by forecast range. Short = 1–7 days, middle = 14–120 days, long = 180–360 days.

Model	Short Avg. RMSE	Middle Avg. RMSE	Long Avg. RMSE	Overall Avg. RMSE
S_Mamba	5.49	14.21	33.13	16.04
iTransformer	5.75	14.34	33.04	16.15
Autoformer	36.06	37.70	43.93	38.71
Transformer	69.11	66.78	74.80	69.56
Informer	79.27	75.60	72.96	76.17
Reformer	68.27	77.47	84.87	76.25

for S_Mamba and 16.15 for iTransformer. The plain attention baselines are substantially weaker on GLD under the locked shared setting, with overall average RMSE above 69 for Transformer and above 76 for Informer and Reformer.

The MAE pattern is similar. S_Mamba and iTransformer remain nearly tied overall, and S_Mamba holds a slightly lower average MAE, 11.61 versus 11.65. These baseline results identify S_Mamba and iTransformer as the two reference models that NS-Mamba must improve upon.

7.1 NS-Mamba Diagnostic Ablation

The first NS-Mamba diagnostic sweep is complete for horizons 30, 120, 180, and 360 days. This is a partial ablation rather than the final 12-horizon result table. The completed diagnostic stage contains 16 finished runs, corresponding to four NS-Mamba variants evaluated on four horizons under the same locked Setting A used by the GLD baselines.

The strongest diagnostic variant is NS-Mamba-D, which uses series decomposition with the S_Mamba normalization pathway and does not use adaptive normalization or shift-aware fusion. Against the better of S_Mamba and iTransformer at each diagnostic horizon, NS-Mamba-D reduces RMSE by 13.12% at 30 days, 8.27% at 120 days, 9.34% at 180 days, and 4.33% at 360 days. Its long-horizon average RMSE is 26.74, compared with 28.90 for S_Mamba and 28.69 for iTransformer. This corresponds to a 7.48% reduction versus S_Mamba and a 6.80% reduction versus iTransformer. The long-horizon MAE also improves, with NS-Mamba-D reducing average MAE by 6.22% versus S_Mamba and 5.50% versus iTransformer.

The diagnostic result is also informative because the more complex variants do not improve the

Table 3: Diagnostic GLD RMSE results for NS-Mamba variants. The long average is computed over 120, 180, and 360 days.

Model	Adaptive Norm	Decomp.	Gate	30	120	180	360	Long Avg.
S_Mamba	No	No	No	11.12	20.31	26.91	39.48	28.90
iTransformer	Yes	No	No	11.45	19.89	26.71	39.45	28.69
NS-Mamba-A	Yes	No	No	11.16	21.16	28.40	43.93	31.16
NS-Mamba-D	No	Yes	No	**9.66**	**18.25**	**24.22**	**37.74**	**26.74**
NS-Mamba-AD	Yes	Yes	No	10.46	20.50	27.49	42.39	30.13
NS-Mamba-ADG	Yes	Yes	Yes	10.64	20.19	27.60	45.50	31.10

Table 4: Direct RMSE and MAE comparison of NS-Mamba-D against all completed GLD models on the overlapping horizons 30, 120, 180, and 360 days. Long Avg. is computed over 120, 180, and 360 days.

Track	Model	30	120	180	360	Long Avg. RMSE	Long Avg. MAE
Opt	PatchTST	9.65	15.32	21.23	33.33	23.29	16.94
Opt	NLinear	9.45	17.53	24.09	34.31	25.31	18.08
Opt	TimesNet	9.40	16.33	24.81	36.63	25.92	18.90
NS-Diag	**NS-Mamba-D**	9.66	18.25	24.22	37.74	26.74	19.54
Shared	iTransformer	11.45	19.89	26.71	39.45	28.69	20.68
Shared	S_Mamba	11.12	20.31	26.91	39.48	28.90	20.84
Opt	Non-stationary Transformer	9.63	20.40	26.76	39.85	29.00	19.99
NS-Diag	NS-Mamba-AD	10.46	20.50	27.49	42.39	30.13	22.34
NS-Diag	NS-Mamba-ADG	10.64	20.19	27.60	45.50	31.10	23.16
NS-Diag	NS-Mamba-A	11.16	21.16	28.40	43.93	31.16	22.97
Opt	DLinear	12.93	27.53	35.23	51.14	37.97	27.13
Shared	Autoformer	37.96	41.02	42.99	44.35	42.79	29.01
Shared	Transformer	66.85	66.08	69.12	81.22	72.14	59.64
Shared	Informer	77.45	72.34	68.18	78.79	73.11	56.60
Shared	Reformer	73.82	81.33	85.29	83.92	83.51	67.99

metric. Adaptive normalization alone is weaker than S_Mamba on the diagnostic long horizons, and adding adaptive normalization to the decomposition branch weakens the decomposition-only result. The shift-aware gate also does not help in this first diagnostic sweep. Therefore, the next experiment should promote NS-Mamba-D to the full 12-horizon GLD evaluation before adding further complexity.

7.2 NS-Mamba-D vs All Completed Models

Table 4 compares NS-Mamba-D against every model that currently has completed GLD results on the overlapping horizons 30, 120, 180, and 360 days. This includes the shared locked-setting baselines, the other NS-Mamba diagnostic variants, and the tuned imported baselines.

NS-Mamba-D ranks 5th at 30 days, 4th at 120 days, 3th at 180 days, and 4th at 360 days among all completed models. On the long-horizon average over 120, 180, and 360 days, NS-Mamba-D ranks 4th overall with RMSE 26.74 and MAE 19.54. It beats every shared locked-setting baseline and every other NS-Mamba diagnostic variant, but it trails the tuned imported PatchTST, NLinear, and TimesNet baselines. This means the decomposition-only NS-Mamba variant is the strongest Mamba-based result currently available in this repository, but it is not yet the strongest model overall on GLD once tuned imported baselines are included.

Table 5: Completed source-optimized GLD coverage for the imported models. Autoformer and Transformer source-style sweeps are still pending.

Model	Sweep rows	Best rows	Max horizon
PatchTST	108	12	360
NLinear	108	12	360
TimesNet	108	12	360
Non-stationary Transformer	108	12	360
DLinear	108	12	360

Table 6: Best completed source-optimized imported results by horizon, measured with RMSE and paired with MAE.

Horizon	Winner	Best seq_len	RMSE	MAE
1	NLinear	320	2.56	1.70
3	NLinear	320	3.53	2.36
5	NLinear	384	4.20	2.85
7	NLinear	192	4.83	3.30
14	NLinear	192	6.54	4.55
30	TimesNet	256	9.40	6.89
60	PatchTST	320	13.34	9.76
90	TimesNet	320	13.29	10.23
120	PatchTST	256	15.32	11.61
180	PatchTST	320	21.23	15.86
270	PatchTST	128	28.13	20.23
360	PatchTST	128	33.33	23.36

7.3 Source-Optimized Imported Baselines

We also ran a separate source-style optimized sweep for imported comparison models using the hyperparameter ranges from the moved `nonstationary_transformer/scripts/GLD.sh` workflow. This sweep is intentionally different from the locked shared-setting baseline table because it uses model-specific hyperparameters and a sequence-length sweep. At the current stage, the optimized sweep is complete for the five imported models Non-stationary Transformer, TimesNet, PatchTST, DLinear, and NLinear.

The completed optimized imported sweep changes the comparison picture substantially. NLinear is the strongest imported model on the short horizons from 1 to 14 days. TimesNet is strongest at 30 and 90 days. PatchTST is strongest from 60 days onward except for 90 days, and it is the best imported model overall under the completed optimized sweep.

The best lookback pattern also depends strongly on the model. NLinear prefers very large windows from 192 to 384. PatchTST prefers 16 on short horizons, then 256 to 320 at middle horizons, and 128 at 270 and 360 days. TimesNet prefers 16 on short horizons and much larger windows at middle and long horizons. Non-stationary Transformer still prefers mostly 16, which appears too short for strong long-horizon GLD performance.

Compared with the best locked-setting baseline at each completed horizon, the optimized imported baselines are substantially stronger except for DLinear. For example, optimized NLinear reduces RMSE by 42.41% at 1 day, optimized TimesNet reduces RMSE by 20.36% at 90 days, optimized PatchTST reduces RMSE by 23.02% at 120 days, and optimized PatchTST reduces RMSE by 15.51% at 360 days. DLinear is weaker than the best locked-setting baseline at every horizon,

while Non-stationary Transformer is competitive at short horizons but no longer competitive with the strongest imported models once the full optimized sweep is complete.

7.4 Complete RMSE and MAE Record

Table 7 lists every completed GLD result currently available in the repository. It combines the locked shared-setting baselines, the NS-Mamba diagnostic ablations, and the best completed source-optimized imported baselines. Only completed results are shown. The current table includes the finished source-optimized imported results for Non-stationary Transformer, TimesNet, PatchTST, DLinear, and NLinear.

Table 7: Current completed GLD RMSE and MAE results across all experiment tracks. For source-optimized imported baselines, the reported sequence length is the best completed `seq_len` for that horizon.

Track	Model	Horizon	Seq. Len	RMSE	MAE
Shared	Autoformer	1	360	32.94	21.44
Shared	Autoformer	3	360	36.21	23.87
Shared	Autoformer	5	360	36.79	24.40
Shared	Autoformer	7	360	38.32	26.17
Shared	Autoformer	14	360	32.35	22.86
Shared	Autoformer	30	360	37.96	26.09
Shared	Autoformer	60	360	38.85	26.83
Shared	Autoformer	90	360	38.32	26.29
Shared	Autoformer	120	360	41.02	28.16
Shared	Autoformer	180	360	42.99	29.21
Shared	Autoformer	270	360	44.47	29.91
Shared	Autoformer	360	360	44.35	29.65
Shared	Informer	1	360	79.69	53.61
Shared	Informer	3	360	78.52	51.75
Shared	Informer	5	360	78.75	53.52
Shared	Informer	7	360	80.14	54.52
Shared	Informer	14	360	77.84	51.90
Shared	Informer	30	360	77.45	52.95
Shared	Informer	60	360	75.53	52.46
Shared	Informer	90	360	74.84	55.13
Shared	Informer	120	360	72.34	53.84
Shared	Informer	180	360	68.18	51.65
Shared	Informer	270	360	71.91	57.90
Shared	Informer	360	360	78.79	64.32
Shared	Reformer	1	360	67.14	42.00
Shared	Reformer	3	360	68.85	43.30
Shared	Reformer	5	360	68.09	44.19
Shared	Reformer	7	360	68.99	46.30
Shared	Reformer	14	360	72.38	51.38
Shared	Reformer	30	360	73.82	53.52
Shared	Reformer	60	360	78.83	59.19
Shared	Reformer	90	360	81.00	62.40
Shared	Reformer	120	360	81.33	63.16
Shared	Reformer	180	360	85.29	69.42
Shared	Reformer	270	360	85.40	71.71

Table 7: Current completed GLD RMSE and MAE results across all experiment tracks (continued).

Track	Model	Horizon	Seq. Len	RMSE	MAE
Shared	Reformer	360	360	83.92	71.39
Shared	S_Mamba	1	360	4.45	3.22
Shared	S_Mamba	3	360	5.30	3.81
Shared	S_Mamba	5	360	5.76	4.14
Shared	S_Mamba	7	360	6.47	4.62
Shared	S_Mamba	14	360	8.17	5.86
Shared	S_Mamba	30	360	11.12	8.09
Shared	S_Mamba	60	360	14.80	10.91
Shared	S_Mamba	90	360	16.68	12.73
Shared	S_Mamba	120	360	20.31	15.28
Shared	S_Mamba	180	360	26.91	19.76
Shared	S_Mamba	270	360	33.00	23.42
Shared	S_Mamba	360	360	39.48	27.48
Shared	Transformer	1	360	67.80	41.90
Shared	Transformer	3	360	70.43	45.08
Shared	Transformer	5	360	68.87	44.19
Shared	Transformer	7	360	69.35	44.63
Shared	Transformer	14	360	68.66	44.63
Shared	Transformer	30	360	66.85	44.89
Shared	Transformer	60	360	66.01	46.74
Shared	Transformer	90	360	66.32	50.12
Shared	Transformer	120	360	66.08	53.08
Shared	Transformer	180	360	69.12	57.69
Shared	Transformer	270	360	74.06	62.18
Shared	Transformer	360	360	81.22	68.14
Shared	iTransformer	1	360	4.59	3.32
Shared	iTransformer	3	360	5.45	3.91
Shared	iTransformer	5	360	6.12	4.35
Shared	iTransformer	7	360	6.85	4.86
Shared	iTransformer	14	360	8.55	6.14
Shared	iTransformer	30	360	11.45	8.23
Shared	iTransformer	60	360	14.89	10.77
Shared	iTransformer	90	360	16.89	12.69
Shared	iTransformer	120	360	19.89	14.87
Shared	iTransformer	180	360	26.71	19.54
Shared	iTransformer	270	360	32.95	23.53
Shared	iTransformer	360	360	39.45	27.63
NS-Diag	NS-Mamba-A	30	360	11.16	8.10
NS-Diag	NS-Mamba-A	120	360	21.16	15.96
NS-Diag	NS-Mamba-A	180	360	28.40	21.09
NS-Diag	NS-Mamba-A	360	360	43.93	31.87
NS-Diag	NS-Mamba-AD	30	360	10.46	7.59
NS-Diag	NS-Mamba-AD	120	360	20.50	15.62
NS-Diag	NS-Mamba-AD	180	360	27.49	20.57
NS-Diag	NS-Mamba-AD	360	360	42.39	30.82
NS-Diag	NS-Mamba-ADG	30	360	10.64	7.66
NS-Diag	NS-Mamba-ADG	120	360	20.19	15.47
NS-Diag	NS-Mamba-ADG	180	360	27.60	20.80
NS-Diag	NS-Mamba-ADG	360	360	45.50	33.21

Table 7: Current completed GLD RMSE and MAE results across all experiment tracks (continued).

Track	Model	Horizon	Seq. Len	RMSE	MAE
NS-Diag	NS-Mamba-D	30	360	9.66	7.18
NS-Diag	NS-Mamba-D	120	360	18.25	13.85
NS-Diag	NS-Mamba-D	180	360	24.22	18.00
NS-Diag	NS-Mamba-D	360	360	37.74	26.78
Opt	DLinear	1	16	4.89	3.43
Opt	DLinear	3	16	5.93	4.41
Opt	DLinear	5	16	6.31	4.39
Opt	DLinear	7	16	7.74	5.67
Opt	DLinear	14	16	9.21	6.56
Opt	DLinear	30	16	12.93	8.76
Opt	DLinear	60	192	18.36	12.84
Opt	DLinear	90	32	23.68	16.61
Opt	DLinear	120	192	27.53	19.77
Opt	DLinear	180	320	35.23	25.53
Opt	DLinear	270	192	45.37	32.78
Opt	DLinear	360	256	51.14	36.08
Opt	NLinear	1	320	2.56	1.70
Opt	NLinear	3	320	3.53	2.36
Opt	NLinear	5	384	4.20	2.85
Opt	NLinear	7	192	4.83	3.30
Opt	NLinear	14	192	6.54	4.55
Opt	NLinear	30	384	9.45	6.45
Opt	NLinear	60	384	13.78	9.33
Opt	NLinear	90	384	16.12	11.24
Opt	NLinear	120	320	17.53	12.97
Opt	NLinear	180	384	24.09	17.20
Opt	NLinear	270	320	29.19	20.56
Opt	NLinear	360	320	34.31	24.08
Opt	Non-stationary Transformer	1	16	2.64	1.79
Opt	Non-stationary Transformer	3	16	4.04	2.69
Opt	Non-stationary Transformer	5	16	4.68	3.17
Opt	Non-stationary Transformer	7	16	5.33	3.61
Opt	Non-stationary Transformer	14	64	6.77	4.75
Opt	Non-stationary Transformer	30	16	9.63	6.55
Opt	Non-stationary Transformer	60	16	14.74	9.70
Opt	Non-stationary Transformer	90	64	18.63	12.52
Opt	Non-stationary Transformer	120	16	20.40	14.43
Opt	Non-stationary Transformer	180	16	26.76	18.74
Opt	Non-stationary Transformer	270	16	32.79	22.58
Opt	Non-stationary Transformer	360	16	39.85	26.81
Opt	PatchTST	1	16	2.58	1.73
Opt	PatchTST	3	16	3.55	2.40
Opt	PatchTST	5	16	4.28	2.92
Opt	PatchTST	7	16	4.91	3.34
Opt	PatchTST	14	16	6.75	4.59
Opt	PatchTST	30	320	9.65	6.75
Opt	PatchTST	60	320	13.34	9.76
Opt	PatchTST	90	256	14.43	10.66
Opt	PatchTST	120	256	15.32	11.61

Table 7: Current completed GLD RMSE and MAE results across all experiment tracks (continued).

Track	Model	Horizon	Seq. Len	RMSE	MAE
Opt	PatchTST	180	320	21.23	15.86
Opt	PatchTST	270	128	28.13	20.23
Opt	PatchTST	360	128	33.33	23.36
Opt	TimesNet	1	16	2.67	1.80
Opt	TimesNet	3	16	3.64	2.46
Opt	TimesNet	5	16	4.35	2.97
Opt	TimesNet	7	16	4.97	3.38
Opt	TimesNet	14	16	6.77	4.64
Opt	TimesNet	30	256	9.40	6.89
Opt	TimesNet	60	384	13.60	10.22
Opt	TimesNet	90	320	13.29	10.23
Opt	TimesNet	120	256	16.33	12.57
Opt	TimesNet	180	192	24.81	18.58
Opt	TimesNet	270	256	30.28	22.27
Opt	TimesNet	360	96	36.63	25.55

There are several plausible reasons why Autoformer, Transformer, Informer, and Reformer are much weaker than S_Mamba and iTransformer on this GLD task. First, the two strongest models use an inverted embedding scheme that maps the full historical window into variable tokens before forecasting, while the four weaker baselines use the standard time-step sequence embedding with encoder-decoder style generation. For a single-target task with a long lookback window, the inverted formulation appears to be better aligned with direct horizon projection than the standard decoder-based setup. Second, S_Mamba and iTransformer both apply per-window mean and variance normalization with final de-normalization, which is likely important for GLD because the price level and volatility change substantially across time. Transformer, Informer, and Reformer in this repository do not apply the same reversible normalization path, and Autoformer applies decomposition but not the same explicit nonstationary normalization. Third, the baseline comparison deliberately uses one shared setting selected around S_Mamba, so the result should be interpreted as a controlled comparison under one configuration rather than as a per-model hyperparameter optimum. Even with that caveat, the size and consistency of the gap suggest that architecture-data mismatch, not only tuning, is a major factor. The completed source-optimized imported sweep reinforces this point from the opposite direction: once model-specific hyperparameters are allowed, PatchTST, NLinear, and TimesNet all become much stronger than their locked-setting counterparts, which means GLD is unusually sensitive to lookback and model-scale choices.

8 Ablation Study

The ablation study isolates the contribution of each NS-Mamba component. The current diagnostic comparison is:

Model	Adaptive Norm	Decomposition	Shift Gate
S_Mamba	No	No	No
NS-Mamba-A	Yes	No	No
NS-Mamba-D	No	Yes	No
NS-Mamba-AD	Yes	Yes	No
NS-Mamba-ADG	Yes	Yes	Yes

The diagnostic result supports the decomposition hypothesis but does not support the adaptive-normalization or shift-gating additions in their current form. NS-Mamba-D is the only variant that improves over both S_Mamba and iTransformer on the diagnostic long-horizon average, and it is also the strongest Mamba-based result currently available in this repository on the overlapping comparison horizons. The full 12-horizon run is still required before making a final claim about the model family.

9 Discussion and Limitations

This manuscript is currently a draft after the baseline stage and the first NS-Mamba diagnostic stage. The main limitation is no longer baseline coverage; that part is complete. The main remaining limitation is that the best diagnostic variant, NS-Mamba-D, has not yet been evaluated across all 12 GLD horizons. The source-optimized imported baseline sweep is complete for PatchTST, NLinear, TimesNet, Non-stationary Transformer, and DLinear, but the optional source-style Autoformer and Transformer sweeps are still pending.

The second limitation is dataset scope. GLD provides a useful nonstationary financial testbed, but one dataset is not enough to establish general forecasting superiority. After GLD experiments are complete, the method should also be tested on standard long-horizon forecasting datasets such as Exchange, Weather, ETT, Electricity, and Traffic.

The third limitation is split sensitivity. Financial time series evaluation can change substantially with different historical periods. The final manuscript should report the exact date range, train-validation-test boundaries, random seeds, and whether validation or test windows use lookback context from earlier splits.

Finally, the architecture may add complexity. The first diagnostic ablation suggests that not every nonstationary module is useful in this setting. Adaptive normalization and shift-aware fusion underperform the simpler decomposition-only branch, so future changes should be justified by improved metrics rather than by architectural novelty alone. The completed baseline table also raises the bar for improvement: NS-Mamba-D now needs to preserve its long-horizon advantage while remaining competitive with S_Mamba at short and middle horizons. The completed optimized imported sweep raises the bar further, because NLinear is strongest at the short horizons and PatchTST is strongest overall under the imported-model tuning regime. The baseline analysis also suggests where the improvement must come from. If NS-Mamba only adds complexity without preserving the inverted-token and reversible-normalization advantages already present in the two strongest shared-setting baselines, it is unlikely to close the gap to the stronger tuned imported baselines for the right reason. The proposed model therefore needs to build on the strengths of S_Mamba while explicitly targeting the long-horizon regime where iTransformer is strongest under the shared setting and PatchTST is strongest under the tuned imported setting.

10 Conclusion

This manuscript presents a structured path for nonstationarity-aware Mamba forecasting. The completed GLD baseline study shows that S_Mamba is the strongest short- and middle-horizon baseline, while iTransformer is the strongest long-horizon RMSE baseline under the shared locked setting. The completed source-optimized imported sweep further shows that NLinear is strongest at short horizons and PatchTST is strongest overall among the tuned imported baselines. The proposed NS-Mamba architecture family combines adaptive normalization, decomposition, and shift-aware fusion around an S_Mamba backbone. The first diagnostic ablation identifies the decomposition-only variant, NS-Mamba-D, as the strongest candidate, reducing diagnostic long-horizon RMSE by 7.48% versus S_Mamba and 6.80% versus iTransformer. The next step is to evaluate NS-Mamba-D across the full 12-horizon GLD suite. The manuscript will become submission-ready after the proposed-model results, ablations, and broader benchmark evaluation are completed with traceable quantitative evidence.

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